

Markscheme

Mathematics: analysis and approaches
Practice question bank

160. (a)

Using De Moivre's theorem, it can be shown that $\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$.

Using this identity, find the value of $\cos 4\theta$ when $\cos \theta = 0.6$.

$$\cos^2 \theta = 0.36, \cos^4 \theta = 0.1296 \quad (M1)$$

$$\cos 4\theta = 8(0.1296) - 8(0.36) + 1 \quad (M1)$$

$$= 1.0368 - 2.88 + 1 \quad \mathbf{A1}$$

$$= -0.8432 \quad \mathbf{A1}$$

[6 marks]